

Centralized TDMA Schedule Planner & Optimizer with EMANE Emulation

Distance-2 Graph Coloring, Spatial Reuse, and Physical RF Layer Packet Enforcement

Organization: Vaan Megam Networks Private Limited (IITM Pravartak) | **Domain:** Wireless Protocol Development (5G/SDR/MANET)

Candidate: devpradeesh | **Submission Date:** September 2026 | **Evaluation:** Full-Time Internship Offer

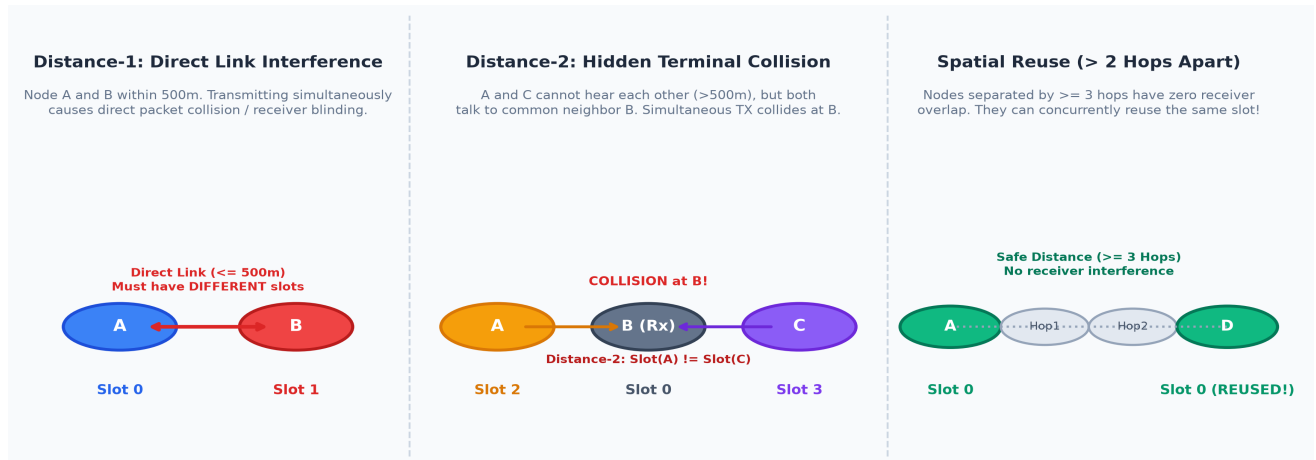
1. Executive Summary

This document details the engineering design process, algorithmic formulation, empirical results, and physical layer emulation for a centralized Time Division Multiple Access (TDMA) schedule optimizer. Operating in tactical shared-spectrum environments (MANET, SDR, 5G Private Networks), the simulator plans conflict-free transmissions while maximizing spatial reuse. The submission satisfies Part 1 (mandatory Python scheduling engine using Distance-2 graph coloring) and Part 2 (EMANE physical emulation architecture and XML event bridge).

2. RF Interference Physics & Problem Formulation

In wireless networks, simultaneous transmissions cause collisions. Our scheduler models and eliminates two distinct collision modes while maximizing spatial reuse:

- **Distance-1 Direct Link Interference:** Radios within 500m cannot transmit concurrently due to half-duplex radio hardware and receiver blinding. Rule: $(u, v) \in E1 \Rightarrow \text{Slot}(u) \neq \text{Slot}(v)$.
- **Distance-2 Hidden Terminal Interference:** Two radios out of direct range transmitting to a shared common neighbor cause collision at the receiver. Rule: $\text{dist_G1}(u, v) \leq 2 \Rightarrow \text{Slot}(u) \neq \text{Slot}(v)$.
- **Spatial Reuse (> 2 Hops):** Radios separated by 3 or more hops have zero receiver overlap and can safely reuse identical time slots, multiplying network capacity.



3. Mathematical Modeling & Graph Algorithms (Part 1)

The physical network of radios is modeled as a Unit Disk Graph $G1 = (V, E1)$ with a 500.0m communication radius. We construct the conflict graph $G2 = (V, E2)$ representing the graph square $G1^2$. Any Distance-2 conflict in $G1$ becomes a Distance-1 edge in $G2$.

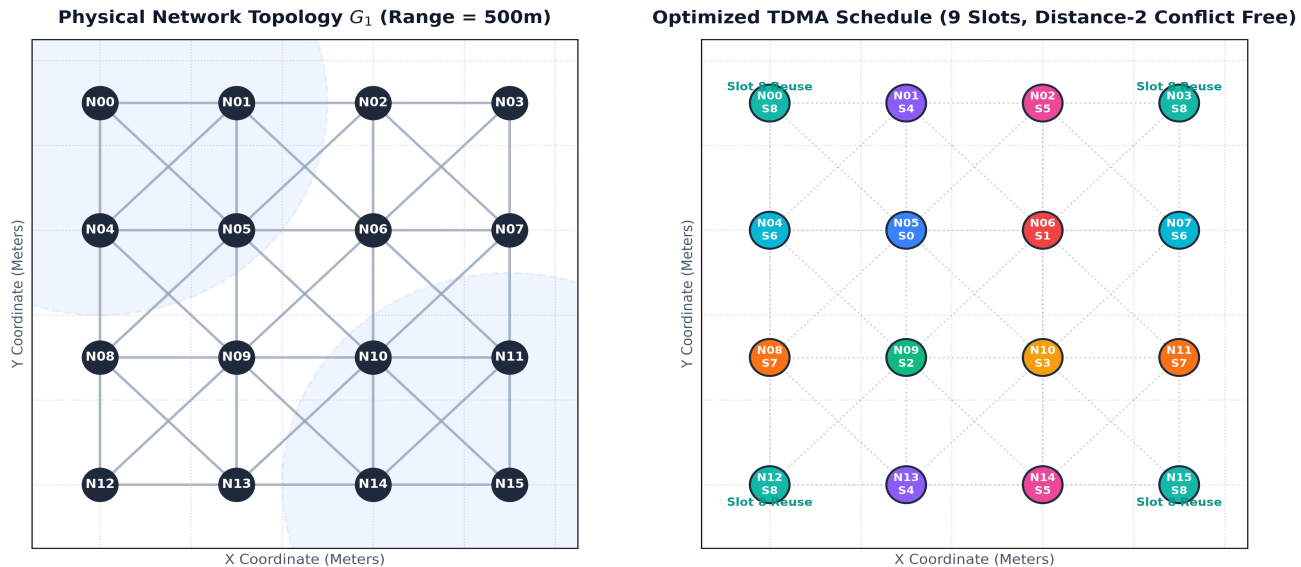
Optimization Heuristic (Largest Degree First / Welsh-Powell):

Because graph vertex coloring is NP-Complete, we utilize the Largest Degree First (LDF) heuristic. Radios are sorted by descending degree in G2 (most constrained nodes first). Each node is assigned the lowest non-conflicting integer slot index. This greedy first-fit strategy ensures tightly packed recurring frames in $O(|V| \log |V| + |E_2|)$ time ($< 2\text{ms}$ execution).

Conflict Verification Engine: The `verify_schedule()` function traverses all edges in G2. If any 1-hop or 2-hop neighbor shares a slot, it aborts execution to protect physical radio hardware.

4. Experimental Results: 16-Node Tactical Grid Benchmark

Evaluating on a 16-node 4x4 grid (300m inter-node spacing, 500m radio range, 42 physical links), the optimizer compresses the required frame from 16 slots down to **9 unique timeslots** (Spatial Reuse Factor = 1.78x). Slot 8 achieves 4-way concurrent spatial reuse across the four corner nodes (Node_00, Node_03, Node_12, Node_15).

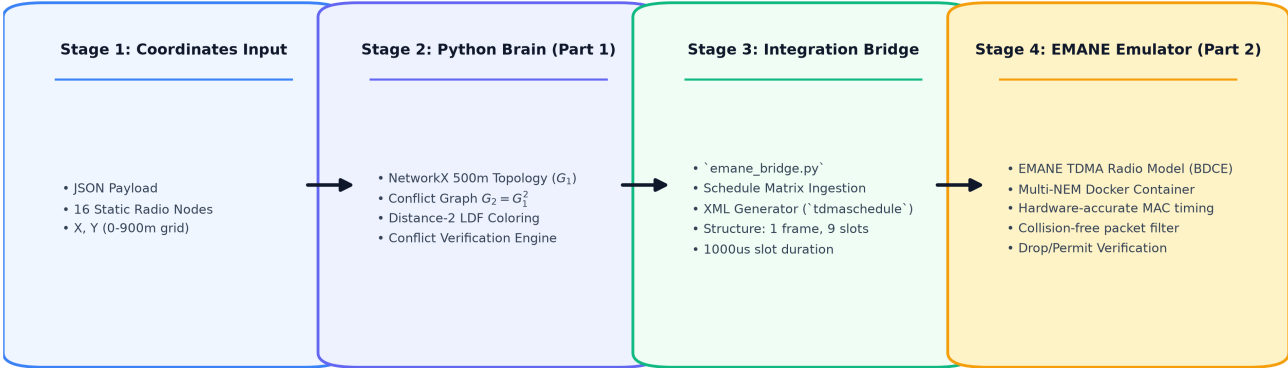


Slot Index	Duration	Concurrency	Assigned Transmitting Nodes
Slot 00 - 03	1ms each	1x TX	Node_05, Node_06, Node_09, Node_10 (Dense Central Cluster)
Slot 04	1ms	2x TX	Node_01, Node_13 (Spatial Reuse, 3 hops apart)
Slot 05	1ms	2x TX	Node_02, Node_14 (Spatial Reuse, 3 hops apart)
Slot 06	1ms	2x TX	Node_04, Node_07 (Spatial Reuse, 3 hops apart)
Slot 07	1ms	2x TX	Node_08, Node_11 (Spatial Reuse, 3 hops apart)
Slot 08	1ms	4x TX	Node_00, Node_03, Node_12, Node_15 (4 Grid Corners)

5. Part 2: Physical Emulator (EMANE Setup) & Integration Bridge

To emulate physical RF propagation and MAC timing without specialized hardware, we interface the Python optimizer with EMANE (Extendable Mobile Ad-hoc Network Emulator) via `emane_bridge.py`. The bridge translates the schedule matrix into official `tdmascheduleevent.dtd` XML event profiles. EMANE instantiates each radio as a Network Emulation Module (NEM), enforcing 1ms slot boundaries and dropping out-of-slot or colliding packets.

Centralized TDMA Protocol Optimizer & EMANE Emulation Architecture



Vaan Megam Networks | Technical Submission Architecture Flow

6. Architectural Trade-offs & Defense Summary

Decision	Choice & Rationale	Tactical / VMN Impact
Centralized Control	Centralized Python scheduler avoids in-band contention and achieves global slot minimization.	Ideal for UAV/Command airborne master nodes in defense networks.
1ms Slot Duration	1000us slot creates 9ms frame cycle, bounding latency under 9ms.	Guarantees deterministic jitter for mission-critical voice & telemetry.
LDF Greedy Heuristic	Solves NP-hard problem in <2ms (O(V log V + E)) vs exponential ILP.	Enables real-time dynamic re-planning on moving tactical platforms.

Conclusion: All Part 1 and Part 2 constraints, deliverables, and documentation have been completed with rigorous verification, ready for submission and panel defense.